

Lecture 2

03/08/24 10:05 AM

Counter example which violates axioms:

Let Ω be $\mathbb{R}$.

$$\mathbb{P}(\Omega) = \mathbb{P}(\mathbb{R}) = 1$$

$\mathcal{P}(\Omega)$ is not always a suitable domain. sometimes violates axioms

Consider unit length intervals

$$[0, 1), [1, 2), \dots$$

$$\mathbb{R} = \bigcup_{n=-\infty}^{\infty} [n, n+1)$$

countable union

$$\begin{aligned} \text{If } \mathbb{P}([n, n+1)) &= x, \text{ then } \mathbb{P}(\mathbb{R}) = \infty \\ \text{"} &= 0, \text{ then } \mathbb{P}(\mathbb{R}) = 0 \end{aligned}$$

$\therefore$ Restrict your domain to only measurable sets.

Possible domain instead of $\mathcal{P}(\Omega)$?

ans: $\{\emptyset, \Omega, \mathbb{R}_-, \mathbb{R}_+\}$

$$\mathbb{P}(\mathbb{R}_-) = \mathbb{P}(\mathbb{R}_+) = 1/2$$

The domain should have some nice properties:

SIGMA-ALGEBRA nice domain for $\mathbb{P}$

The event space or sigma-algebra $\mathcal{F}$ associated with a set Ω is a collection of subsets of Ω that satisfy:

- $\emptyset \in \mathcal{F}$ and $\Omega \in \mathcal{F}$
- $A \in \mathcal{F} \Rightarrow A^c \in \mathcal{F}$
- $A_1, A_2, \dots, A_n, \dots \in \mathcal{F} \Rightarrow \bigcup_{n=1}^{\infty} A_n \in \mathcal{F}$

- σ -algebra is closed under the formation of complements and countable union.
- Also closed under formation of countable intersections. (since union belongs and complement belongs, apply de Morgan's law).

* When Ω is countable and finite, $\mathcal{P}(\Omega)$ is a sigma-algebra.

* For any set $S \in \Omega$, the smallest sigma-algebra containing S is $\{\emptyset, \Omega, S, S^c\}$

FORMAL DEFINITION of probability measure $\mathbb{P}$:

A probability measure $\mathbb{P}$ on the measurable space $(\Omega, \mathcal{F})$ is a function $\mathbb{P}: \mathcal{F} \rightarrow [0, 1]$ s.t.

- $\mathbb{P}(\emptyset) = 0, \mathbb{P}(\Omega) = 1$
- $\mathbb{P}(\bigcup_{i=1}^{\infty} A_i) = \sum \mathbb{P}(A_i)$
countable additivity

The trio $(\mathbb{P}, \Omega, \mathcal{F})$ = probability space

Probability space for $U[0, 1]$:

uniform

$$\Omega = [0, 1]$$

$$\mathcal{F} = \{\emptyset, [0, 1], [0, 0.5], [0.5, 1]\}$$

But we know $\mathbb{P}([0.25, 0.75]) = 0.5$

If we add this set, we have to add many more sets to satisfy σ -algebra properties. Eventually we have to include all kinds of interval brackets.

$$[], (), [), (]$$

BOREL-SIGMA ALGEBRA

- When $\Omega = [0, 1]$, the $\mathcal{B}[0, 1]$ is the σ -algebra generated by closed sets of the form $[a, b]$ where $a \leq b$, and $a, b \in [0, 1]$

$$\text{Since } (a, b) = \bigcup_{n=1}^{\infty} [a + \frac{1}{n}, b - \frac{1}{n}] \text{ and } (a, b] = \bigcap_{n=1}^{\infty} (a, b + \frac{1}{n}]$$

$\mathcal{B}[0, 1]$ is the σ -algebra generated by sets of the form $[a, b], (a, b), (a, b]$ or $[a, b)$ where $a \leq b$ and $a, b \in [0, 1]$.

- If $\Omega = \mathbb{R}$, then $\mathcal{B}(\mathbb{R})$ is the σ -algebra generated by open sets of the form (a, b) , where $a \leq b$ and $a, b \in \mathbb{R}$.

Q: How would you define $\mathcal{B}(\mathbb{R}^2)$?

CONSEQUENCES OF THE PROBABILITY AXIOMS proved using probability axioms

$$\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$$

A, A^c are disjoint. $A \cup A^c = \mathbb{R}$. Apply countable additivity

$$\begin{aligned} \mathbb{P}(A \cup B) &= \mathbb{P}(A \setminus B) + \mathbb{P}(B \setminus A) + \mathbb{P}(A \cap B) + \mathbb{P}(A \cap B) - \mathbb{P}(A \cap B) \\ &= \mathbb{P}(A \setminus B) + \mathbb{P}(A \cap B) + \mathbb{P}(B \setminus A) - \mathbb{P}(A \cap B) + \mathbb{P}(A \cap B) \\ &= \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B) \end{aligned}$$

$$\begin{aligned} \text{If } A \subseteq B, \quad \mathbb{P}(A) &\leq \mathbb{P}(B) \\ \mathbb{P}(B) &= \mathbb{P}(A) + \mathbb{P}(B \setminus A) \end{aligned}$$

Q: Prove: $\mathbb{P}(\bigcup_{i=1}^{\infty} B_i) \leq \sum_{i=1}^{\infty} \mathbb{P}(B_i)$ Boole / Bonferroni's inequality

Q: Prove inclusion-exclusion principle.

Zero probability $\neq$ Impossible

$$\mathbb{P}(\emptyset) = 0 \rightarrow \text{impossible}$$

LIMITS AND CONTINUITY refer NCERT

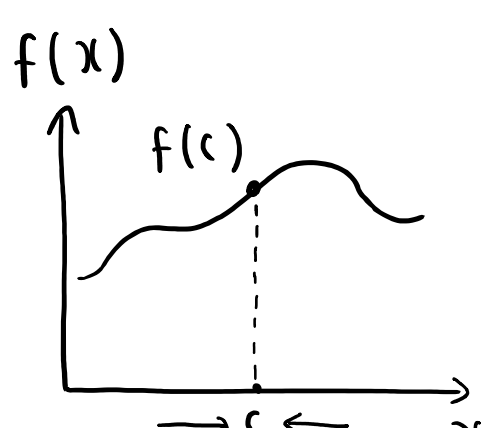
Limit of a sequence $\{a_1, a_2, \dots\}$

$$\lim_{n \rightarrow \infty} a_n = L$$

(ϵ, N) definition

Limit of a function at a point c

$$\lim_{x \rightarrow c} f(x) = L$$



$$\text{Continuity} \Rightarrow LHL = RHL = f(x)$$